

How Small Businesses Transform PDF Agreements into Action

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ABSTRACT

A legal agreement is a type of procedural document that describes the steps that parties must take to fulfill legal obligations. Following these steps requires human interpretation, which is often inefficient and error prone. For a Small to Medium Sized Business (SMB), this process is laborious. We conduct an interview study to understand how information in PDF agreements is currently understood, processed, and acted upon by SMB employees working in small teams of non-domain experts. Through qualitative analysis and a text highlighting activity, we observe knowledge transfer workflows in SMBs and propose design principles for using AI-extracted information to create actionable documents that address gaps in efficiency, understanding, and agency.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Empirical studies in collaborative and social computing.**

KEYWORDS

PDF; extraction; agreement visualization; actionable documents; small business

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1 INTRODUCTION

Legal agreements define binding obligations that parties must fulfill through various actions, making them a type of procedural document [5]. In contrast to other traditional procedural documents, such as technical documentation or assembly instructions,

agreements commonly require the participation of multiple parties and span longer term time periods of months or years. Identifying obligations within legal agreements requires careful collaborative human interpretation, which can often be laborious, repetitive, and error prone. For readers without legal expertise, completing those actions is especially confusing, as legal agreements have low contract clarity, a measure of usability for non-legal professionals [6, 7], and are often subject to task-based reading [5], in which readers prioritize information most relevant to their immediate goals. Work around contract clarity focuses on visualization to match this gap, such as textured agreements with distinctive text and graphics [9], charts of common characteristics across document structures and possibility spaces [21], agreement comics [22], and more expansive forms of multi-modal law visualization that use audio-visual components [2], though such methods are applied manually based on document context and not scalable.

There has been significant exploration in making different documents and data types interactive using AI tools. While AI is not adept at automating context-specific information, it can be powerful for system customization by providing a lower-level structure that can be shaped by a user to fit a higher-level need [11], providing an opportunity to use AI to shape the human processing of legal agreements through extraction. Document intelligence research focuses on layered document annotation [17] [12], visualization of embedded events in agreements [8] [15] [4], surfacing key information [16, 19], human-in-the-loop information extraction [18], and pairing generated summaries with highlighted source text to increase trust [14], though the impact of transparency on trustworthiness varies [20]. Existing research largely focuses on crystallizing understanding through the analysis and curation of documents, while less attention has been paid to employing these techniques to facilitate taking action from documents.

Accordingly, with a high-level goal of examining action in procedural documents, we ask: **How might we help non-domain experts fulfill legal obligations in PDF agreements using AI-extracted information?** To understand pain points in current PDF agreement to action workflows, we begin by interviewing Small to Medium Sized Business (SMB) employees who work with PDF agreements, as they are often working in teams of non-domain experts on end-to-end processes of decentralized agreement workflows. SMBs in particular face barriers in accessing enterprise tools to process PDF agreements at scale and maintaining digital literacy to navigate business needs [10]. While agreements contain many clauses and action types, we focus our investigation on payment

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clauses in service agreements due to their quantifiable implications and recurring responsibilities. Prior work around payment automation revolves around smart contracts [13, 23] and programmable money [1, 3]. We report on our findings from an interview study with SMB employees. Specifically, we make the following contributions:

- An investigation on how information in PDF agreements is transformed into business actions by SMB teams
- Design principles for interfaces that aim to visualize AI-extracted information and support agreement workflows

2 METHODS

To answer our research question, we conducted an interview study with SMB employees (Table 1). The goal of this study was to understand participants' pain points around taking action on agreements with a focus on payment actions. We explored both the granular details of manual information extraction from a singular agreement clause, as well as the high-level workflows in moving from agreements to payment processes.

We interviewed 11 U.S.-based SMB employees (6 female, 5 male), who were recruited through purposeful sampling. Our recruitment criteria required that participants 1) worked in businesses that had 1-100 employees and 2) had familiarity with both payment clauses in agreements and payment workflows within their business. We recruited participants through UserTesting.com and customer mailing lists. As a result, all participants were SMB employees, experienced agreement readers, and confident in reading agreements. Participants were compensated with gift cards.

Hour-long interviews were conducted online via a videoconferencing tool. In the first part of the interview, we asked background questions to understand participants' legal expertise and responsibilities. The second part of the interview focused on understanding participants' current workflow around agreements, how they reach their own understanding of agreement obligations, and how they work with others to take action on these obligations. Finally, to observe how participants manually extract information and to understand the utility of automated extraction, we asked participants to read a clause and review additional extracted information (Paying Party, Receiving Party, Amount Due, Pay Date(s), and Trigger) from the clause. Participants were then asked to extract information from the clause below using a Qualtrics survey:

"If Borrower makes monthly payments in advance, 1 percent interest will be deducted by Lender starting on the 18th day of the next month after the advance payment, with a maximum total deduction of 6 percent interest, leaving the minimum amount of interest possible at 5 percent for all monthly payments."

3 FINDINGS

To analyze our findings, we conducted thematic analysis on the interview data. We organize common findings, across participants' industries and roles, into three overarching themes: 1) Identifying Immediate Obligations through Trigger-Action Pairs, 2) Organizing Action through Decentralized Tools, and 3) Taking Action through Repetitive Communication.

3.1 Identifying Immediate Obligations through Trigger-Action Pairs

In identifying obligations, all participants shared that they understand obligations as "Trigger-Action" pairs. Despite displaying strong confidence in understanding individual obligations, participants shared struggles with contextualizing obligations within the full scope of an agreement and defining sub-details of each obligation.

3.1.1 Trigger-Action Mental Model. All participants displayed a mental model that distilled each obligation, or "anywhere where there's language where somebody shall pay or shall do something" as P2 put it, into a **Trigger-Action pair**: a **trigger**, an event that makes an obligation relevant, and an **action**, the task needed to fulfill the obligation itself. Participants used different language to describe Trigger-Action pairs, such as "*call to action*" (P5), "*cause and effect*" (P9), or "*if and then*" (P8). This "Trigger-Action" mental model is a user-defined framework for document understanding, though it is still subject to individual interpretation. Codifying this process may help unify reader processing, provide further structure in agreement review, and extract obligations at scale.

3.1.2 Myopic Understanding of Trigger-Action Pairs within Agreement Context. While participants understand obligations as Trigger-Action pairs, understanding the relationship between multiple obligations is difficult, since "*there is the possibility of infinite lawsuits*" (P4). As shown through contracts represented as DFA graphs [4], agreements cover many Trigger-Action pairs, though not all will be realized. The "happy path", or the required obligations to fulfill an agreement, changes based on the reality of the project. Participants shared that the labor of tracking relevant trigger events often outweighs the benefit of doing so, leading them to prioritize obligations by immediacy, as P7 says:

"Tracking all these actions is almost wasted time— I don't want to deal with things that I don't need to do today."

This highlights the myopic task-based nature of agreement review, as SMB employees prioritize the action they need to take in the short-term. When discussing client management, participants shared the difficulty of referencing multiple obligations within the context of one agreement. P6 described this experience from her calls, saying that she is "*just sitting there going digitally through the pages, looking for the part that [the client] is talking about*". Participants need more comprehensive understanding of agreement scope and a way to temporally and semantically search within them.

3.1.3 Conflicts in Confidence when Identifying Trigger-Action Sub-details. We also observed that defining sub-details in Trigger-Action pairs often requires reader interpretation, while experienced readers' confidence in a clause might overshadow their confidence in a sub-detail, leading to potential misunderstanding of terms. Participants were shown a portion of the clause that contained a Trigger-Action pair and asked to identify extraction details, which mirrored details shown in the additional information provided for other clauses in the study: Paying Party, Receiving Party, Amount Due, Pay Date(s), and Trigger. Participants faced some difficulty in determining what each extraction detail should map to (Table

Table 1: Participant Demographics

Number	Age Range	Business Size (Employees)	Gender	Industry	Role
P1	40-49	5	Male	Art	Artist, Creative Director
P2	50-59	1	Female	Legal	Lawyer
P3	30-39	20	Female	Design Consulting	Chief Operations Officer
P4	50-59	1	Female	Pet Care	Owner, Caretaker
P5	50-59	30	Male	Consulting	CEO
P6	40-49	44	Female	Office Supplies	Manager
P7	50-59	10	Male	Housing	Landlord
P8	30-39	50	Male	Education	Financial Manager
P9	20-29	20	Male	Consulting	Consultant
P10	20-29	2	Female	Beauty	CEO
P11	30-39	7	Female	Sports Technology	Contract Manager

2). Some participants looked back at the example extraction details highlighted to determine their own highlights, while others created personal definitions of each one. Some participants also wanted to add additional fields to mark, since they valued some information that did not necessarily fit into a detail. P10 shared that defining a sub-detail is difficult in regards to due dates that are dependent on other dates:

"[for dates] like when the contract was signed or "semi-annually" . . . it would be nice to get an exact date, but if not, I would just have to do a manual type of calculation which I wouldn't really like to do."

3.2 Organizing Action through Decentralized Tools

After discussing their individual review and extraction of agreement information, participants shared their process in managing and fulfilling obligations concerning payment. Currently, the PDF format of legal agreements does not match the reality of a project, so understanding is developed and documented through other document formats.

3.2.1 Decentralized Payment Obligation Workflow and Lack of Motivation. Participants shared their laborious and disorganized workflows in moving from signing a PDF agreement to managing a payment process. Though all participants managed payment processes closely, they did not view it as a task that they wanted to spend time doing. P2 said, "*I should be lawyering, not collecting debt*", while P1 shared "*Payment is annoying. It's not part of my creative process*". On average, participants described using a minimum of 3 tools in a payment process, including email to communicate payment terms, personalized spreadsheets to keep personal notes on client situations, color-coded folders to represent payment status, project management systems, and payment management systems. One participant described her manual tracking through local folders as "*archaic*" (P11), while another participant depended on clients "*sending [files] to [her] email so it's searchable*" (P4).

3.2.2 Moving from Agreements to Payment through non-PDF Documents. In taking action to fulfill obligations, all participants used

non-PDF documents to track obligations, document their understanding, and take action. Participants described the original PDF as a document that solely provides legal protection as a ground truth, instead of one that actively provides value throughout a process. Once an agreement is finalized, a participant identifies the information most relevant to their situation and only revisits that PDF if the situation changes or they need to double-check terms. Instead of referencing a PDF agreement, participants described creating documents to represent their understanding of the agreement that can be updated to match the status of a project. P3, the Chief Operations Officer of a design consulting agency, describes this process as the following:

"The agreement is signed and then the client pays us a deposit and we start work on it. The Scope of Work (SOW) is sent to a program manager. The program manager uses that to compile a deck that outlines the deliverable for the project, so the key people are interacting with the SOW deck. That SOW is also used to build what is called the Project Register. The Project Register outlines the timeline."

Currently, this translation of agreement terms is entirely dependent on the documentation of a team member, increasing the potential of incorrectly documenting details as agreements get longer and more convoluted.

3.3 Taking Action through Repetitive Communication

In taking action on identified obligations after trigger events occur, SMB employees must communicate internally with their own team, which depends on hierarchical decision-making, and externally with their clients, which often involves repetitive communication of the original agreement terms.

3.3.1 Taking Action within SMB Team. In taking action with an SMB team, once a participant is satisfied with the personal documentation they have made for an agreement and trigger events concerning payment occur, they are ready to proactively create

Table 2: Highlighting Activity Results

Subdetail	Highlighted Text
Trigger	"If Borrower makes monthly payments in advance" "makes monthly payments in advance" "monthly payments in advance" "payments in advance"

more official documents such as invoices and receipts that are associated with finality of action. P9 describes the decision-making that leads up to payments being made:

"After [signing a] contract, we still need to do background checks, especially for lawsuits. For doctors that could take another month [before we pay them]."

The recurring and final nature of creating these official documents leads to repetitive action for an SMB employee at varied timelines, with P11 stating that she is *"pretty accustomed to almost on a weekly basis sending invoices"* based on sending an *"invoice once per season, sometimes twice per season"* per customer. All participants used a separate third party tool to handle invoicing, including Excel, checks, QuickBooks, Stripe, and LawPay.

3.3.2 Taking Action with Clients. In communicating with clients to take action, SMB employees must prioritize information on behalf of a client and reiterate terms multiple times to ensure the client knows why they have an obligation and how to fulfill it, which is often accomplished through repetitive and frustrating communication. P2, a lawyer running her own firm, described the frustration that can come with clients misunderstanding payment terms and her self-defined work timelines, saying that:

"If I stop working, they get angry. Once I send a package out and the application has been accepted, my work is done. It's always specifically laid out in the agreement. I try to put as much information in there."

This common misunderstanding leads her to communicate with clients around terms repeatedly in detail—she drafts the agreement itself, calls the client to go over the agreement, sends them an overview with agreement terms in simplified language, sends a Jira spreadsheet to maintain knowledge, then regularly reminds them of upcoming obligations based on nearing trigger events. Each of these reminders reiterates language from the original agreement, which is another instance of repetitive labor in the payment process.

4 DESIGN PRINCIPLES

Our findings show that for SMB employees across various industries and roles, legal agreements are expanded procedural instructions subject to task-based review, in which instructions are dynamically updated based on trigger events, the procedure is long term, and obligations require the action of multiple parties. There is a need for structure, centralization, and decreased repetitive labor in moving from agreements to payment. **To aid non-domain experts**

in fulfilling legal obligations in PDF agreements, we propose the following design principles for using AI-extracted information in actionable document interfaces:

- **Provide structure for reviewing and documenting Trigger-Action pairs for obligations within an agreement.** Manually extracting obligations from an agreement is laborious and error prone (Section 3.2.1), as details are often subject to reader interpretation (Section 3.1.3). Guide user understanding of obligations by defining sub-details and showcasing templates for users to define sub-details themselves.
- **Dynamically visualize Trigger-Action pairs to match the status of a project.** Agreements cover many potential situations (Section 3.1.2), so visualize Trigger-Action pairs according to project status for more proactive understanding and easier reference. Provide modules that allow users to prioritize, sort, navigate, and understand such obligations by details such as due date and effective date.
- **Allow flexible use and human understanding of extracted information.** Participants re-purpose building blocks of extracted information in different formats throughout a payment process (Section 3.2.2 and 3.3.1). Allow users to easily turn extracted information into a usable form throughout a process, whether as notes, documentation, receipts, invoices, or other documents.

5 DISCUSSION AND CONCLUSION

Future research will explore implementing these design principles in actionable document prototypes, matching reader mental models around clause extraction, and establishing trust with automated documents in the legal domain. This will require larger scale studies around clause reading, extraction review, and actionable document prototype testing with both expert readers, such as lawyers, and non-expert readers, such as SMB employees with no legal background. Future work can expand beyond the use case of SMB employees working with payment clauses to understand different knowledge transfer workflows. This also inspires future direction around technical functionality, such as relative date extraction and classifying extracted information into Trigger-Action pairs. Finally, this research calls for intentional research around reader trust of AI extraction tools in the legal domain generally. Applying AI in the legal domain carries ethical responsibility to maintain original context around extracted information, practice transparency on how information is analyzed, and enable flexibility to match user preferences in how information is identified.

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